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Paleomagnetism of the Miocene Jantetelco Granodiorites and Tepexco Volcanic Group and inferences for crustal block rotations in central Mexico — Reevaluation

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ABSTRACT

Miocene igneous rocks from a location in central Mexico provide paleomagnetic data that allow re-evaluation of previous data by Urrutia-Fucugauchi (1981) that were interpreted to suggest a counter clockwise rotation of a crustal block by about 50°. We sampled 31 sites from the Miocene trondhjemitic Chalcatzingo domes, the Tepexco volcanic group, and the Xalostoc diorite, covering the same area of approximately 50 km^2 about 90 km SE of Mexico City. Magnetic analysis shows that these rocks contain magnetic phases of variable composition, with Curie temperatures characteristic for magnetite, but often accompanied by lower Curie temperature components. Magnetic hysteresis measurements point to the presence of PSD particles, and in combination these properties suggest that these rocks are also suitable for paleomagnetic study. Scanning electron microscope analysis supports the rock magnetic results, indicating the presence of magnetite and high-intermediate titanium titanomagnetite in many samples. Demagnetization experiments showed in most cases characteristic remanence directions of reasonable to good quality, and for 26 sites mean directions could be determined. Of these 14 (12) are of normal (reverse) polarity. Overall mean directions of normal and reverse sites are antipodal and pass a reversal test at the 95% probability level. The paleodirection from 26 sites is D = 348.1°E, I = 35.7°, α_{95} = 7.4°, and the paleopole is located at Lat = 78.3°N, Long = 184.2°E, A_{95} = 7.0°, which is indistinguishable from the 20 Ma reference pole for the stable North America plate. These data do not support any tectonic deformation of the sampling area since the Miocene. Based on the number of sites studied, their rock magnetic characteristics, and the quality of the magnetic remanence that results in a positive reversal test, we consider our data to be reliable and we therefore suggest that this result supersedes that of Urrutia-Fucugauchi (1981). The reason for this previous and much different result remains unknown.

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1. Introduction

Paleomagnetism uses the earth's magnetic field recorded in rocks as a marker for kinematic analysis of movements in the crust. It has been successfully applied for reconstructing tectonic processes on very different scales. One of the first applications was to test the hypothesis of continental drift, by comparing paleopoles from different tectonic plates (e.g., Collinson and Runcorn, 1960; Creer et al., 1959; Irving and Green, 1958; Runcorn, 1956). On a smaller scale, the displacement of tectonostratigraphic terranes has been studied and it has been suggested that they have traveled over long distances until their amalgamation to another tectonic plate, e.g., for the western North America Cordillera (e.g., Beck, 1980; Irving et al., 1996). Finally, neotectonic processes in active lateral fault systems have been quantified in terms of their extent and temporal evolution using paleomagnetic data to determine crustal block rotations, e.g., in California (Luyendyk et al., 1980) or the Aegean (Kissel and Laj, 1988).

Mexico today belongs to the North America plate, but most of southern Mexico is believed to be a mosaic of allochtonous terranes that were accreted in the past. Baja California is an example for such accretion processes: while it was part of the North America plate before about 12 Ma ago, it later separated by the introduction of the East Pacific Spreading center and the opening of the Gulf of California and accreted to the Pacific plate. Today it moves roughly northward with respect to the North America Plate. Southern Mexico is composed of several terranes that have moved into their present position after breakup of Pangea (e.g., Sedlock et al., 1993) (Fig. 1). Paleomagnetic studies resulting in divergent paleopoles have not only provided evidence for such movements previous to the Cretaceous (e.g., Ballard et al., 1989;

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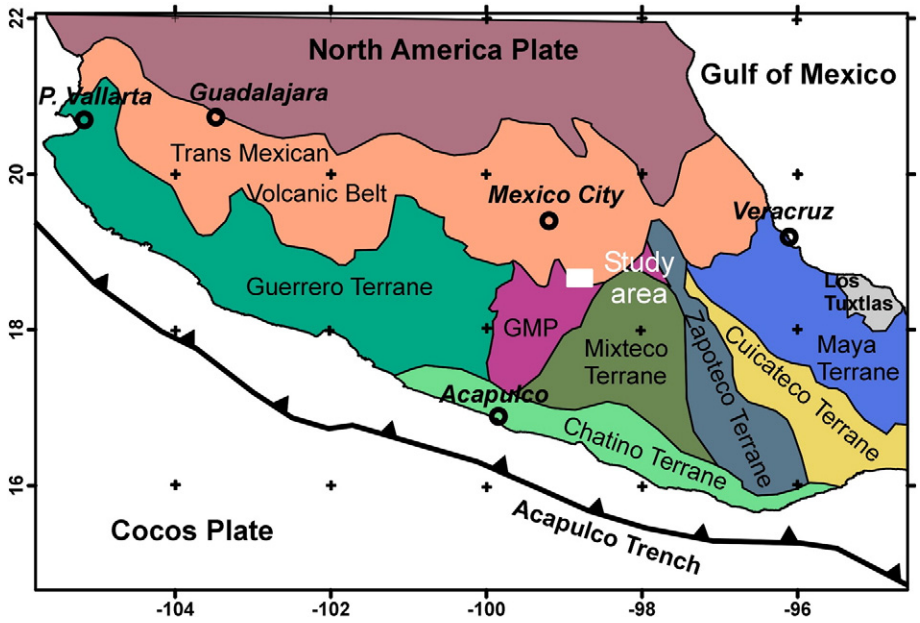


Fig. 1. Terrane map of southern Mexico and the Trans Mexican Volcanic Belt (modified after Ortega-Gutiérrez et al., 2012). GMP, Guerrero-Morelos platform. Major cities are shown with black circles for reference. The study area is marked by a white rectangle.

Böhnel, 1989; Godínez-Urban et al., 2011), but also shown that most of Mexico had acquired its present day configuration probably since the mid-Cretaceous (Böhnel et al., 1989, 2002; Molina-Garza et al., 2014).

The Transmexican Volcanic Belt (TMVB) is a subduction related volcanic arc that has evolved since about 15 Ma (e.g., Aguirre-Díaz et al., 1997; Ferrari et al., 1999). It is situated between several tectonostratigraphic terranes (Fig. 1), and while these are of much older age, these terranes may have influences the origin and orientation of the belt. With respect to other volcanic arcs, the TMVB forms an angle of ~20° with respect to the subduction trench. Three main hypotheses to explain this anomaly are: 1. Variable subduction angle along the trench, decreasing from ~50° in the NW to subhorizontal in the SE (Pardo and Suarez, 1995). This shallowing subduction of the angle would result in a depth of magma fusion that becomes more distant to the subduction trench to the SE. 2. The presence of old suture zones between terranes along the track of the TMVB that facilitate the up rise of magma (e.g., Molnar and Sykes, 1969; Mooser, 1972). 3. A transtensional shear system in response to differences in plate velocities of the North America, Pacific and Caribbean plates (Shurbet and Cebull, 1984). Such a shear system is also considered to be viable in view of the neotectonic activity along the TMVB, as shown by structural geology data and by historic earthquakes that have occurred within the upper crust (e.g., Suter et al., 1995, 2001).

Early paleomagnetic results seemed to support the presence of a left lateral shear system beneath the TMVB (Urrutia-Fucugauchi, 1981; Urrutia-Fucugauchi and Böhnel, 1987, 1988). Paleomagnetic mean directions from 10 rock units of Cretaceous to Quaternary age distributed along the TMVB showed in most cases counter clockwise rotated declinations, but only rarely a significant inclination difference. The data were thus interpreted as indicators of localized crustal block rotations around vertical axes within a left lateral shear system (Urrutia-Fucugauchi and Böhnel, 1987, 1988). Notably, later paleomagnetic studies from the TMVB have rarely encountered similar counter clockwise rotations (e.g., reviews by Mejia et al., 2005; Ruiz et al., 2010), so that the validity of these early paleomagnetic studies may be in doubt. We present a re-evaluation of the first study which proposed such a counter clockwise rotation of a crustal block in central Mexico, by ~50° (Urrutia-Fucugauchi, 1981). This work is based on the thesis results of Vazquez Duarte (2010) and complementary data obtained before and afterwards.

2. Geology of the study area and paleomagnetic sampling

The study area is located southeast of Mexico City, along the central part of the TMVB (Fig. 1). The geology has been studied by Fries (1966) and is resumed by Rivera et al. (1998) in the 1:250,000 geologica map published by the Servicio Geológico Mexicano. More recent results by Gómez-Tuena et al. (2008) and Ortega-Gutiérrez et al. (2012) report the geochemistry and petrology of the Chalcatzingo domes (previously known as Jantetelco granodiorites) and the xenoliths contained in them.

In the study area, the local basement is not exposed but overlain by the Cretaceous Guerrero-Morelos platform (GMP) (Fig. 1), mainly formed by platform-facies carbonate rocks. Located to the west of the GMP is the Jurassic to Cretaceous Guerrero terrane, and to the east the Paleozoic Mixteco terrane, and the basement in the study area could thus correspond to either of these. Based on the analysis of numerous xenoliths occurring in the Chalcatzingo dome, Ortega-Gutiérrez et al. (2012) suggest that the local basement is of Proterozoic age, which excludes the Guerrero and rather favors the Mixteco terrane. The GMP is unconformably overlain by the Eocene Balsas Formation, a sequence of continental conglomerates and red bed sedimentary rocks, which in turn is covered by Oligocene to Miocene volcanic rocks of the Buenavista-Tepoztlan and the Tilzapotla (named by Fries (1966) in eastern part of our study area Tepexco volcanic group) Formations. These rocks are covered by widespread upper Pliocene lahars of the Cuernavaca Formation and alluvial sediments. Intrusive rocks like the Xalostoc diorite are of similar age as the Chalcatzingo domes (Fries, 1966).

Several tectonic processes have affected the study area, with the oldest events of pre-Cenozoic age. These will not be discussed here, as they could not have affected the Miocene igneous rocks we have studied. During the Cenozoic and Quaternary, three tectonic events are important. The oldest and most important event is the Laramide orogeny, which took place between 80 and 40 Ma and was related to increased convergence between the Pacific and North America plates (e.g., Engebretson et al., 1985). This resulted in the Sierra Madre Occidental fold belt and important compressional tectonic features in the Sierra Madre del Sur (Cerca et al., 2007) and thus possibly led to the folding observed in the rocks of the MGP. This deformation pre-dates the emplacement of the Miocene rocks in the study areas.

Later, extension known as Basin and Range tectonics affected most of western North America and Mexico, as far south as Oaxaca (Henry and Aranda-Gomez, 1992). This phase of extension involved normal faulting and may have produced listric block displacement within the mainly N–S oriented graben structures. Basin and range faulting was active since at least 49 Ma (Aranda-Gómez and Mcdowell, 1998) and may have been most active between 23 and 12 Ma (Henry and Aranda-Gomez, 1992). Block tilting in extensional tectonic environments has indeed been reported (e.g., Aranda-Gómez and Mcdowell (1998) in the Guanajuato mining district about 350 km to the NW,) but is not observed in the study area.

The third and most recent process is related to differential plate movements mainly between North America and the Caribbean plates (Guzmán-Speziale, 2001), and is well expressed in the Motagua–Polochic fault system. It has also been proposed that such neotectonic activity observed along the TMVB (e.g., Johnson and Harrison, 1990; Suter et al., 1995) reflects a transtensional shear system.

Close to the study area, no evidence has been reported for major lateral faults (e.g., Johnson and Harrison, 1990; Morán-Zenteno et al., 2007). Morán-Zenteno et al. (2004) suggest an ENE oriented left-lateral shear zone about 70 km south of the study area, with possible activity not later than 32 Ma, well before the emplacement of the Chalcatzingo domes. Finally, Rivera et al. (1988) only show one inferred normal fault with a NE trend, along the NW scarp of an anticline of the Morelos Formation limestone (see Fig. 2). In their geologic map, bedding attitudes for the Pleistocene Cuernavaca Formation are marked for several localities, with low dips of only a few degrees and rather random strike angles. Therefore, the study area has not experienced any significant shortening since the Pleistocene. Together with the well exposed sub-vertical magma flow structures in the Chalcatzingo domes (Ortega-Gutiérrez et al., 2012) we exclude any larger post-emplacement tilt of these bodies.

Gómez-Tuena et al. (2008) studied the geochemistry and petrology of the three domes (Fig. 2) previously known as Jantetelco granodiorites and renamed them as Chacatzingo domes. They include viscous lavas of trondhjemitic composition with an $^{40}Ar/^{39}Ar$ age of 20.7 Ma, which confirms the early Miocene age assignation of Fries (1966). This age estimate is similar to that of the volcaniclastic Tepotzlan Formation which is located about 40 km NW, representing the time interval 22.8–18.8 Ma (Lenhardt et al., 2010). Both the Chalcatzingo domes and the Tepotzlan Formation thus may represent the initial phase of the TMVB. These recently obtained age estimates place these rocks in the Miocene, and not the Oligocene, as indicated by Rivera et al. (1998) in their stratigraphic column.

Paleomagnetic sampling was mainly carried out in the Chalcatzingo domes and the Tilzapotla Formation or Tepexco volcanic group of similar stratigraphic age, thus allowing a meaningful comparison with the previous study by Urrutia-Fucugauchi (1981) in this same area. Additionally, three sites were sampled from the Xalostoc diorite body, which is also of Miocene age (Fries, 1966). All in all, 31 sites were sampled (Fig. 2), of which 25 have precise GPS coordinates; the remaining six sites were sampled time ago without GPS localization, but their location is well known from positions on topographic maps, and they closely relate to some of the former sampling sites of Urrutia-Fucugauchi (1981). Sampling was carried out with a portable drill and 12 or 25 mm drill bits with diamond tips. Drill cores were oriented with an inclinometer and a magnetic compass, and we took care to avoid places where the magnetic compass was deflected due to lightning strike influences. Such effects were more frequent when climbing up from the archeological site towards the summit of the central dome. A total of 285 cores were recovered, giving in average almost 9 cores per site.

Most sites sampled for the present work are not the same as the sites of Urrutia-Fucugauchi (1981), which were easily recognized by their drill holes, but could not be re-sampled during our recent field work, as they were located within the main area of the important Chalcatzingo archeological site. Compared with the previous study, our sampling covers a significantly larger area of about 50 km^2 (Fig. 2).

3. Previous results

Urrutia-Fucugauchi (1981) reported results from nine sites (total of 50 samples), with seven belonging to the Chalcatzingo domes and two to the Tepexco volcanic group. Thermal and alternating field (AF) demagnetization results were interpreted to provide stable site mean directions without any important secondary overprint for the Chalcatzingo domes. Rocks from the Tepexco volcanic group showed important magnetization overprints, but demagnetization was interpreted to successfully eliminate this overprint and resolving stable endpoint directions. Four sites showed reverse and five normal polarity directions, which were approximately antipodal although no statistical test was applied to corroborate this. The combined mean direction calculated was 301.2°E/+27.1°, α_{95} = 10.7°, with the corresponding overall mean paleomagnetic pole at 176.3°E/33.7°N, A_{95} = 9.5°. Comparing

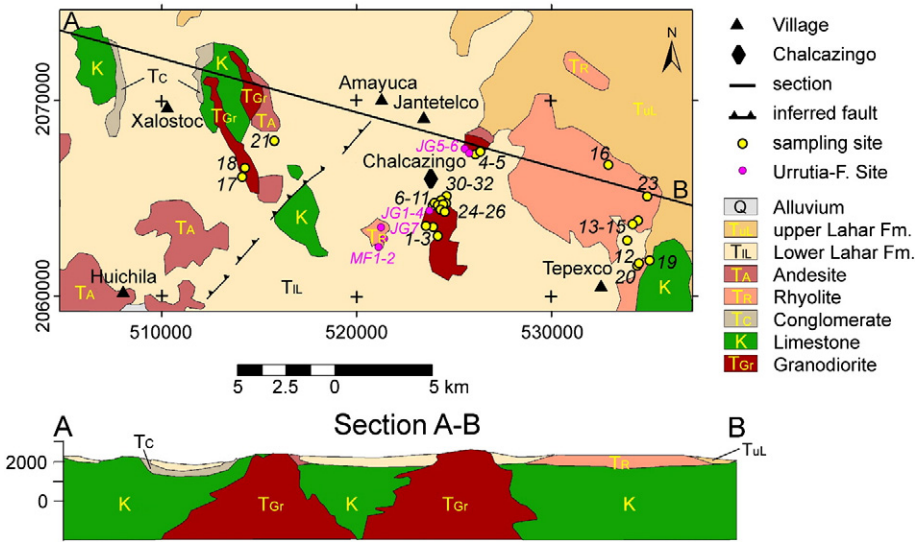


Fig. 2. Geological sketch map of the study area (top), with a vertical profile along line A–B (bottom), and the corresponding stratigraphic column. Modified from Rivera et al. (1998).

these data with reference data from stable North America led the author to propose that the study area had been rotated around a nearby vertical axis by about 50–55° in a counter clockwise sense, possibly within a left lateral shear system. As this result was based on only nine sites, Urrutia-Fucugauchi (1981) mentioned in his conclusions that "*More detailed palaeomagnetic sampling is necessary to give a more complete picture of the tectonic evolution*". Nevertheless, to date this recommendation has not been followed up.

We note that paleomagnetic data from 79 nearby (~40 km) sites of the Miocene Tepotzlan Formation have been reported by Lenhardt et al. (2010), which were successfully used for a magnetostratigraphic age determination of this volcaniclastic sequence. Although part of these collected rocks was not demagnetized and therefore their mean directions are not considered of high quality, the site mean directions yield an antipodal distribution of normal and reversed directions (see Fig. 7 in Lenhardt et al., 2010). As opposed to the results of Urrutia-Fucugauchi (1981), these data cluster roughly around 350°E/30° and thus do not suggest any major tectonic deformation of the area. Farther to the north, in the Mexico-City basin, a similar result is reported by Mooser et al. (1975), who studied a total of 41 sites of upper Oligocene to Pliocene volcanic rocks, with overall mean directions close to those for stable North America, implying a tectonic stability of this area (e.g., Urrutia-Fucugauchi and Böhnel, 1988).

4. New rock magnetic and paleomagnetic results

For characterization of the composition of magnetic minerals, thermomagnetic analysis was carried out on selected samples. For that purpose, up to 400 mg of crushed rock was used in a horizontal Curie balance (Petersen Instruments model VFTB) and heated in air up to 700 °C, in a constant magnetic field of about 500 mT. Typical heating and cooling curves are given in Fig. 3, and show rather variable behavior, even within one rock unit. Curves from the Chalcatzingo domes show an almost reversible thermomagnetic curve with a Curie temperature T_c ~ 560 °C, indicating a titanium-poor titanomagnetite (sample CH8, Fig. 3a). However, site CH26 corresponds to the northern of the two southernmost domes and shows a much different behavior. The heating curve shows two Curie temperatures around 350 °C and 570 °C, and the cooling curve is irreversible with a notable destruction of the low-T_c component. We explain this behavior by the combination of magnetite and titanomagnetite with intermediate titanium content, which oxidized during the heating to higher temperatures and was thus converted into a low- or non-magnetic phase.

The heating curve of the volcanic rock CH15 is similar to the last one, with two Curie temperatures around 370 °C and 570 °C (Fig. 3c), but in this case the cooling curve shows a single T_c of ~580 °C and an important increase in magnetization. Here the low-T_c mineral was converted into magnetite, and we assume that this could either be maghemite or a

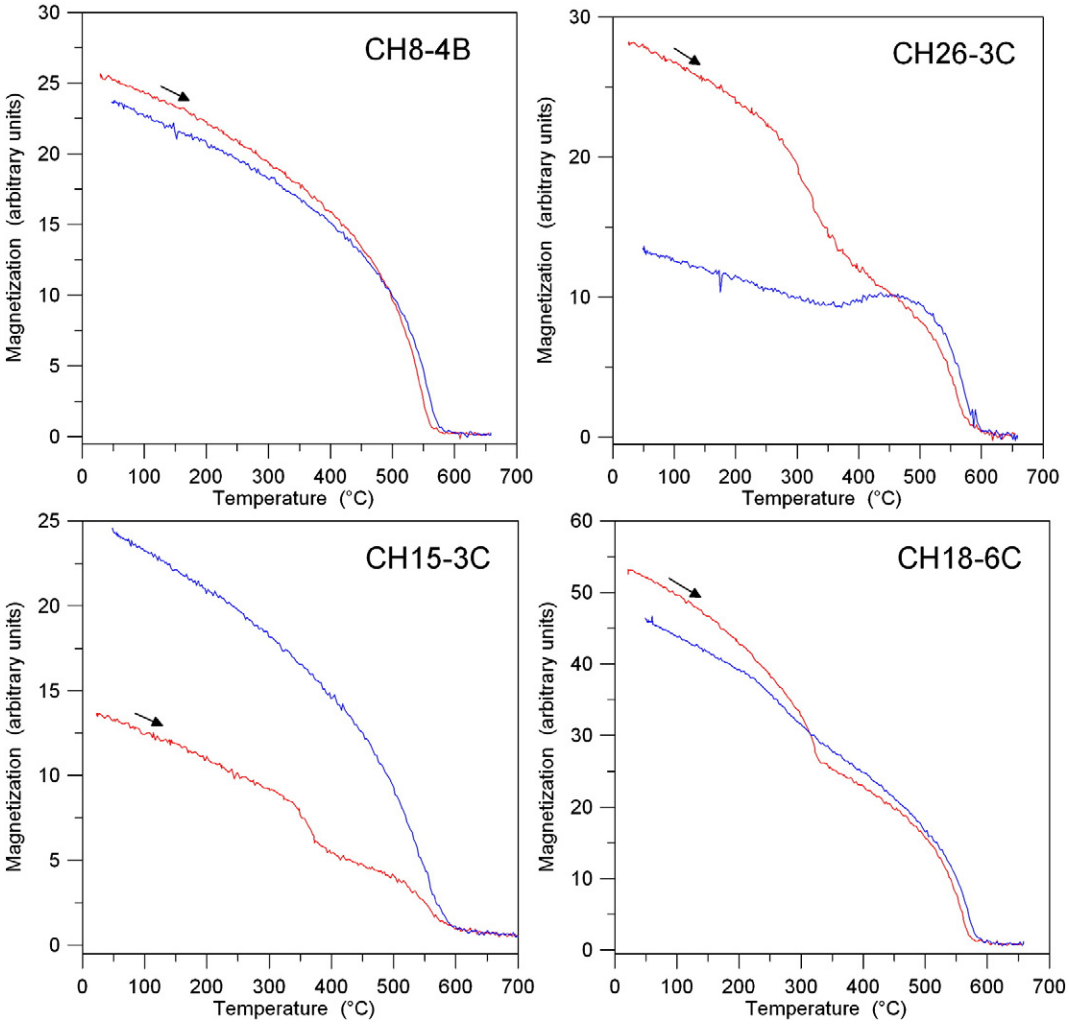


Fig. 3. Thermomagnetic curves of samples from the Chalcatzingo granodiorite (CH8-4B, CH26-3C), Tepexco volcanic group (CH15-3C), and the Xalostoc diorite (CH18-6C). Curves in red (blue) indicate heating (cooling) branches.

sulfide mineral. Finally, site CH18 from the Xalostoc diorite is characterized by a heating curve with two Curie temperatures around 320 °C and 570 °C, and a cooling curve where the low-T_c component is suppressed but still visible, and Curie temperatures around 580 °C and 300 °C. In this rock, we have therefore a dominant contribution of magnetite, and another mineral that is mostly destroyed during heating, such as pyrrhotite or an intermediate-titanium titanomagnetite.

Magnetic hysteresis measurements of rock chips were carried out using a Princeton Measurement Corporation model MicroMag 2900 vibrating sample magnetometer. Typical hysteresis curves are shown in Fig. 4. Many samples are characterized by constricted hysteresis loops and low to intermediate coercive forces of ~10–30 mT (CH2-3 and CH26-3C from the Chalcatzingo domes; CH12-7C and CH14-3 from the Tepexco volcanic group). These curves saturate after correction of a minor dia-/paramagnetic contribution at fields ~300–600 mT, indicating the presence of (titano-) magnetite with some high coercive contribution of variable relative amount. That high coercivity component could be hematite, goethite, or an alteration product of (titano-) magnetite. Another hysteresis loop (CH16-1c, Tepexco volcanic group) shows strong paramagnetic effects and at the same time a relatively high coercive force > 60 mT. After correction of the paramagnetic contribution, this curve is clearly wasp waisted and thus indicating the presence of at least two mineral phases of different compositions and/or grain sizes. Finally, sample CH21-3b (Xalostoc diorite) is dominated by a diamagnetic contribution, which most probably is due to the phenolic sample holder that becomes important because of the small magnetic moment of this rock. Correcting for the diamagnetic contribution, this sample as well has a wasp waisted shape and a coercive force of ~40 mT. All curves are characteristic for a grain size in the range of pseudo single domain particles (PSD).

The rock magnetic studies suggest that the magnetic minerals present in the rocks examined are of variable composition, where magnetite is often accompanied by a mineral of low Curie temperature, possibly titanomagnetite. The magnetic granulometry is a characteristic of PSD particles, which are of high magnetic stability.

These samples were also prepared for observation with a scanning electron microscope (SEM). We used a Hitachi TM1000 with an acceleration voltage of 15 kV and elemental composition was obtained by energy dispersive X-ray spectroscopy (EDS) using an Oxford microanalyzer coupled to the microscope. For SEM analysis a paleomagnetic sample was polished and its surface covered with graphite. In Fig. 5, typical examples of SEM images are shown, and for all of the oxide particles displayed electron diffraction spectra are also available, which indicate their elemental composition in a semi-quantitative way. Note that samples shown in Fig. 5 coincide with those in Fig. 3. All samples are characterized by a large if not dominant contribution of magnetite or titanomagnetite with low titanium content, which is especially true for smaller grains < 20 μm (e.g., Fig. 4b, c, e). Additionally, there are large particles > 30 μm, often with a trellis-type intergrowth of magnetite and high-titanium titanomagnetite or ilmenite (Fig. 5a, f), as well as smaller (< 10 μm) homogeneous high-titanium titanomagnetite particles (Fig. 5b, d). Iron sulfide minerals are observed only very rarely in a sample from the Tepexco intrusive rocks, and thus seem to be of negligible importance for the rock magnetic properties. The SEM analyses support the interpretation of the thermomagnetic measurements, indicating that the magnetic remanence carriers are largely magnetite and titanomagnetite of variable titanium content.

The natural remanent magnetization (NRM) of all samples was analyzed with stepwise demagnetization. For that purpose we used an AGICO model LDM AF demagnetizer and an ASC Scientific model TD48 thermal demagnetizer. AF amplitudes up to 100 mT were used, and for thermal demagnetization temperatures up to 600 °C. Typical demagnetization curves are shown in Fig. 6 as orthogonal vector plots. Both demagnetization methods successfully removed unstable components

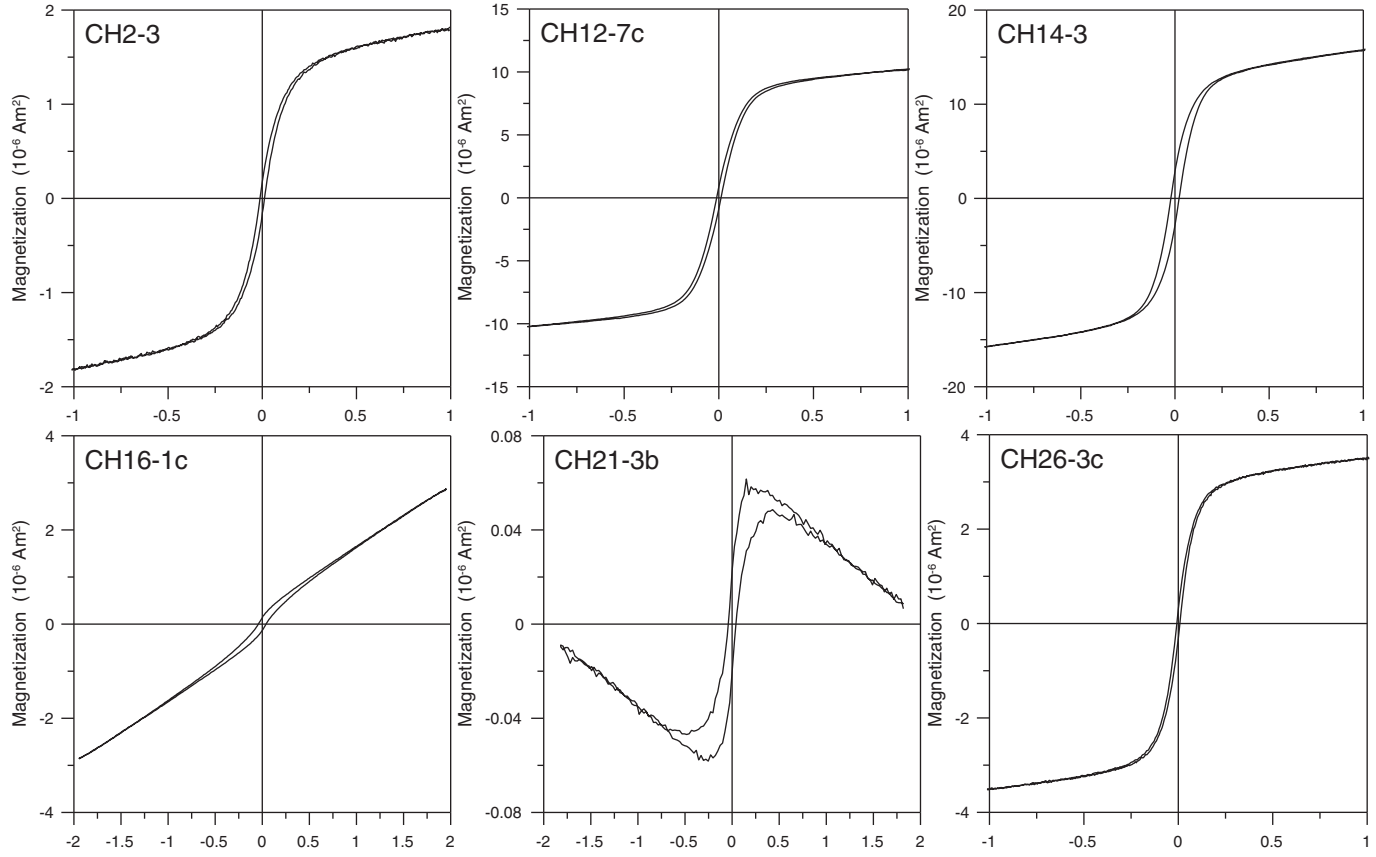


Fig. 4. Magnetic hysteresis curves for samples from different rocks and sites. Note: no correction for dia- and paramagnetic contributions has been applied.

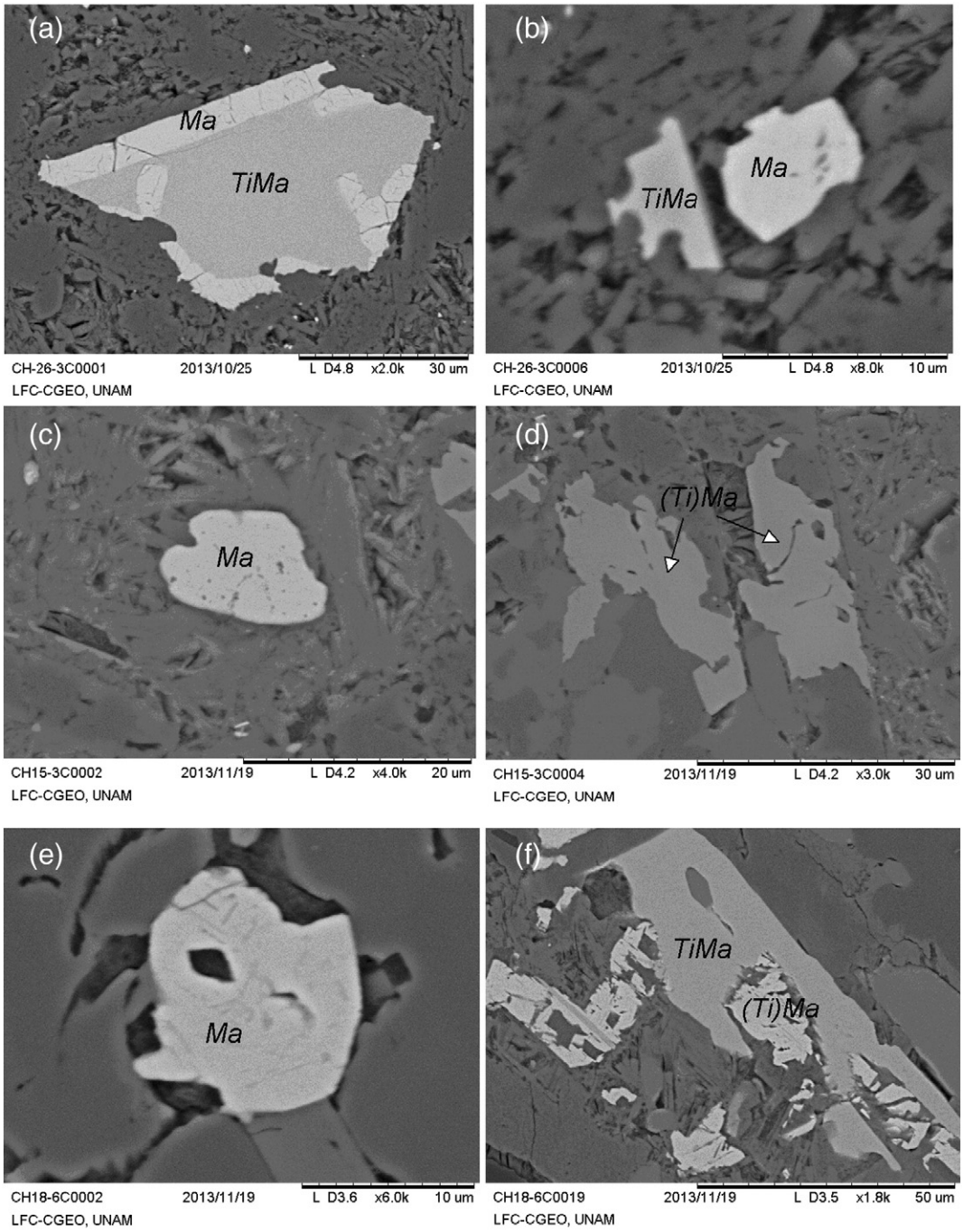


Fig. 5. SEM backscattered electron images for samples from sites CH26, Ch16, and CH18. *Ma*, magnetite, *TiMa*, titanomagnetite, (*Ti*)*Ma*, titanium-poor titanomagnetite. A scale (um, micrometers) is shown for each image in the lower right corner, together with the amplification number.

considered to be of secondary origin during the first treatment steps, and after wards demagnetization curves trended towards the origin, thus indicating the presence of a stable characteristic remanence direction (ChRM). Secondary components are of variable relative contribution, compared to the ChRM: although there are samples with small secondary components (e.g., Fig. 6a, c, f), other samples are more affected by such overprints (Fig. 6b, d, e). This phenomena occurs between sites and also within a single site, as shown by the demagnetization diagrams in Fig. 6a and d, which both correspond to site CH5. Here thermal demagnetization led to the definition of a well defined ChRM with only a small secondary remanence contribution (Fig. 6a), while AF-demagnetization (Fig. 5d) removed a strong overprint of normal direction in field amplitudes up to 30 mT, and after wards isolated the same stable ChRM direction as derived from the thermal experiment. Additional demagnetization diagrams are included in the supplementary data file.

For every sample, a ChRM direction was calculated applying the principal component analysis (PCA) method to the demagnetization data (Kirschvink, 1980). Only those samples with more than three demagnetization steps included in the linear fit and with a maximum angular deviation value MAD < 10° were accepted to be included in a site-mean direction (Table 1). In case of more scattered demagnetization curves or of very high coercivity magnetization, the Fisher (1953) mean direction was used for a demagnetization interval of at least 3 steps, with a maximum $\alpha_{95} = 10°$. These requirements reduced the number of samples per site slightly, compared with the drill cores

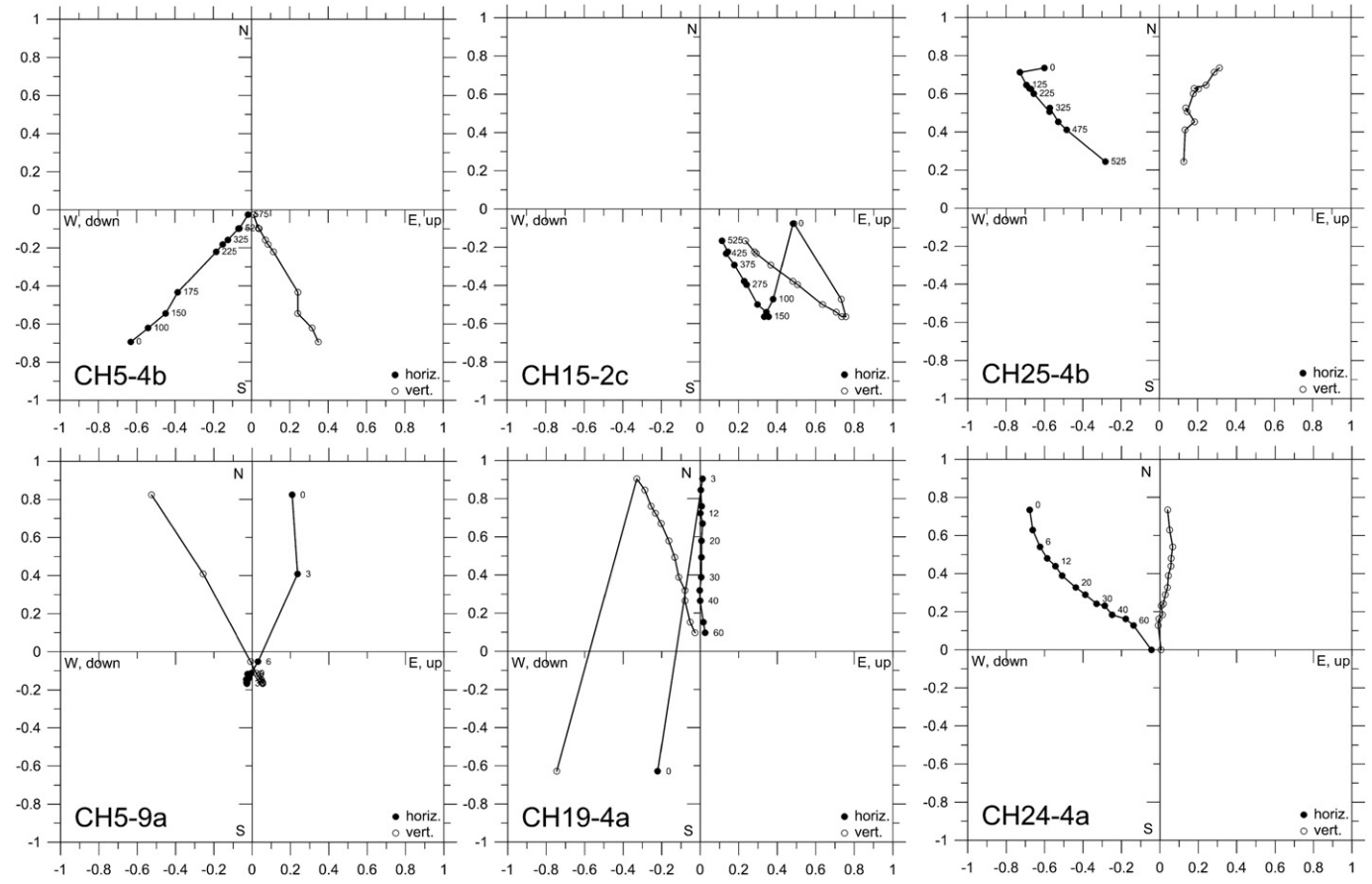


Fig. 6. Orthogonal vector plots from thermal (upper three images) and alternating magnetic field (lower three images) stepwise demagnetization experiments. Labels along the curves are temperatures (°C) or field amplitudes (mT) used for demagnetization.

recovered in the field. In five sites, the number of rejected samples and/or the dispersion of their ChRM directions was too large to be acceptable, and no site-mean direction could be calculated, thus reducing the number of accepted sites to 26. Site-mean directions were calculated from the accepted ChRM directions and are listed in Table 1, together with their statistical parameters and with the corresponding virtual geomagnetic poles (VGP). The site-mean directions and VGPs are also plotted in equal area projection diagrams (Fig. 7).

Site-mean directions are of acceptable quality, with most 95% confidence limits $\alpha_{95} < 10°$ (6 sites have $10.4° < \alpha_{95} < 15.5°$). Fourteen sites are characterized by normal and 12 by reverse site-mean directions. Mean directions were calculated for both polarities (Table 2 and Fig. 7), and these directions pass a reversal test at the 95% probability level. Using the program PMag Tools (Version 4.2b by Mark W. Hounslow) the observed Gamma = 9.48° was found to be smaller than the critical Gamma = 14.72°. The combined overall mean direction based on all 26 sites given in Table 1 is 348.1°/+35.7°, with k = 15.5 and $\alpha_{95} = 7.4°$ (see Table 2). From the site-mean VGPs a paleopole was calculated, situated at 184°E/78.3°N, $A_{95} = 7.0°$.

Paleosecular variation (PSV) of normal and reverse polarity VGPs and of the combined data set was analyzed, and the angular standard deviation s with its upper and lower limits s_u and s_l are listed in Table 3. Values of s are between 17.7° and 19.0° with confidence limits between 14.0° and 26.5° and thus are slightly larger than predicted by PSV model G (McFadden et al., 1988) at the site latitude, s ~ 13.5°. This VGP dispersion together with the positive reversal test suggests that these data have averaged out paleosecular variation and are representative for both field polarities.

5. Discussion

Most of the sites sampled for this study are from the Miocene Chalcatzingo trondhjemitic domes, with some volcanic and intrusive rocks of similar stratigraphic position. These rocks contain magnetic minerals of composition and grain size as often found in igneous rocks, and we consider them to have recorded faithfully the ambient geomagnetic field during their initial cooling; this record has been recovered by stepwise AF and thermal demagnetization experiments. Secondary magnetization components are of variable importance, probably because of the variable effect of lightning induced magnetization within the sampling sites. Site mean directions could therefore be calculated for most of these sites, with α_{95} confidence limits between 2.8° and 15.4°, with 20 of 26 sites having an $\alpha_{95} < 10°$. The site mean directions are of both normal and reverse polarity, with 14 and 12 sites, respectively. Data sets of both polarities averaged out secular variation reasonably well, as they show mean directions (Table 2) that pass a reversal test and PSV is similar as expected for model G. We therefore assume that the overall mean direction and paleomagnetic pole are characteristic for the geologic period these rocks belong to, and may be used for analyzing any tectonic implications. For comparing these data with data from stable North America, we use the reference paleopole determined by Besse and Courtillot (2002) for the time of 20 Ma (Table 3), which is very similar to the geochronologic age of the Chalcatzingo dome of 20.7 Ma. From this reference pole a reference direction was calculated for the study area (Table 2), with an estimated $\alpha_{95} = 4.2°$. This value is an overestimate, as due to the transformation from the reference pole to a direction the confidence limits become

Table 1
Paleomagnetic site mean results. Site code; geographic coordinates from GPS readings; rock unit, CRD = Chalcatzingo rhyolite domes, TVG = Tepexco volcanic group, XD = Xalostoc diorite; N, number of drill cores; n, number of samples used for site mean; R, unit vector sum; k, precision parameter, α_{95}, 95% confidence limit; Dec, mean declination; Inc, mean inclination, Lat, VGP latitude; Long, VGP longitude. Site coordinates marked with a * symbol are only approximate. n.d., not determined.

Site	Latitude	Longitude	Unit	N	n	R	k	α_{95}	Dec	Inc	VGP-Lat	VGP-Long
	°N	°W						(°)	°E	(°)	(°)	°E
CH1	18.66231	98.77113	JGD	10	0				n.d.			
CH2	18.66484	98.77338	JGD	9	9	8.92806	111.2	4.9	331.4	37.5	63.0	180.9
CH3	18.66568	98.77593	JGD	9	8	7.98263	402.9	2.8	358.4	64.1	62.8	258.8
CH4	18.69834	98.75105	JGD	13	12	11.93638	172.9	3.3	192.9	−16.6	−73.8	208.7
CH5	18.69866	98.75036	JGD	9	8	7.80000	35.0	9.5	199.7	−21.6	−69.9	189.9
CH6	18.67459	98.77156	JGD	8	6	5.77697	22.4	14.5	352.8	31.4	82.9	158.2
CH7	18.67484	98.77149	JGD	10	10	9.93363	135.6	4.2	357.0	35.2	87.1	186.1
CH8	18.67524	98.77129	JGD	11	11	10.82175	56.1	6.2	358.8	21.6	82.4	90.1
CH9	18.67546	98.77003	JGD	10	10	9.91313	103.6	4.8	0.6	30.0	87.3	68.7
CH10	18.67544	98.76939	JGD	8	8	7.54248	15.3	14.6	2.0	29.3	86.4	48.6
CH11	18.67373	98.76847	JGD	10	0				n.d.			
CH12	18.64813	98.67221	TVG	11	10	9.87805	73.8	5.7	350.5	24.8	79.2	140.7
CH13	18.66735	98.67361	TVG	7	6	5.87864	41.2	10.6	172.8	−35.8	−83.1	1.8
CH14	18.66671	98.67462	TVG	9	8	7.69432	22.9	11.8	167.3	−27.3	−77.1	333.9
CH15	18.65894	98.67838	TVG	11	10	9.83871	55.8	6.5	144.5	−46.8	−56.2	13.9
CH16	18.69313	98.68663	TVG	10	10	9.80519	46.2	6.8	167.6	−55.1	−69.8	50.8
CH17	18.68800	98.86533	XD	9	0				n.d.			
CH18	18.69165	98.86405	XD	9	9	8.89041	73.0	6.1	179.1	−31.0	−87.8	284.8
CH19	18.65002	98.66672	TVG	10	10				n.d.			
CH20	18.64778	98.67250	TVG	9	9	8.83707	49.1	7.4	354.8	34.2	85.1	172.8
CH21	18.70457	98.85040	XD	8	7	6.90581	63.7	7.6	174.3	−24.6	−82.0	305.4
CH23	18.67883	98.66805	TVG	8	8	7.86916	53.5	7.6	149.1	−61.6	−54.4	40.7
CH24	18.67685	98.76740	JGD	9	8	7.90085	70.6	6.6	315.4	−1.6	42.1	152.2
CH25	18.67713	98.76793	JGD	6	6	4.97775	179.8	5.7	321.9	12.6	50.9	157.8
CH26	18.67682	98.76798	JGD	8	8	5.92754	69.0	8.1	308.4	42.4	41.8	187.8
CH27	18.6990*	98.7550*	JGD	6	6	5.93397	75.7	7.7	172.8	−42.5	−81.1	34.0
CH28	18.6990*	98.7550*	JGD	6	6	5.91844	61.3	8.6	156.5	−49.6	−65.7	24.6
CH29	18.6990*	98.7550*	JGD	6	5	3.93978	49.8	13.1	177.9	−36.8	−87.3	34.0
CH30	18.6990*	98.7550*	JGD	8	6	5.93325	74.9	7.8	358.4	47.6	79.9	253.2
CH31	18.6990*	98.7550*	JGD	7	6	5.79772	19.8	15.4	329.9	35.8	61.6	178.5
CH32	18.6990*	98.7550*	JGD	9	0				n.d.			

non-circular around the reference direction; the value given corresponds to the major axis of the confidence oval and is used in Table 2 and the corresponding plot in Fig. 8. Both mean and reference directions are very similar, with an angular distance of only 5.1°, while the sum of the corresponding confidence limits is 10.1°. In Fig. 8 this similarity is clearly visible, as both directions overlap widely with their confidence limits. Tectonic movement was calculated according to Demarest (1983), giving an apparent rotation of R = −4.9° ± 10.4° and an inclination flattening F = −2.9° ± 7.9°. Considering the error limits, both values are therefore insignificant.

Both the Chalcatzingo paleopole and the 20 Ma reference pole are plotted in Fig. 9, together with their 95% confidence circles. Again it is clear that both poles are very similar, with widely overlapping confidence circles. Applying the calculations proposed by Demarest (1983) for poles (Table 3), a rotation of −5.3° ± 6.4° and a poleward displacement of −3.2° ± 6.0° is obtained, which again does not indicate any

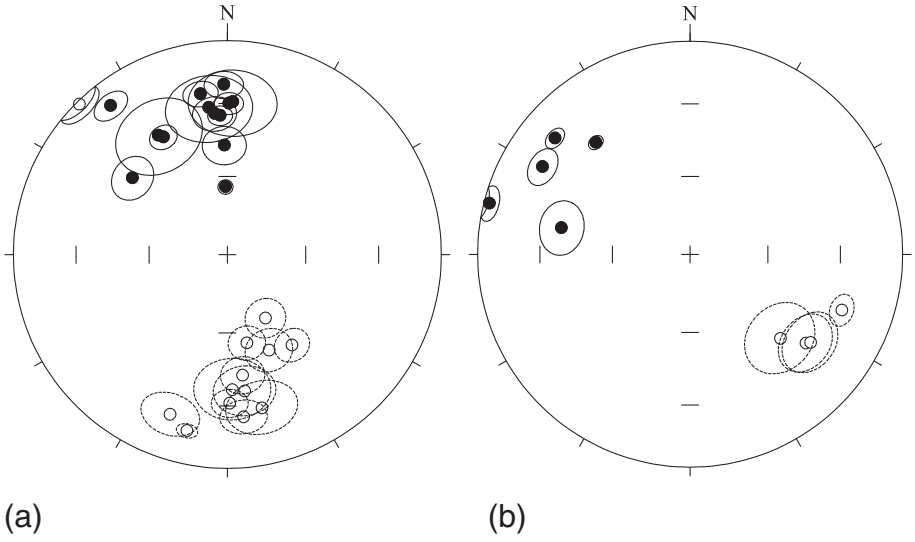


Fig. 7. Site-mean directions with 95% confidence circles in the equal area projection. a, results from this work, b, results from Urrutia-Fucugauchi (1981). Black dots correspond to positive inclination, circles to negative inclinations. Equal area projection, generated with PMag Tools, Version 4.2b.

Table 2
Mean directions obtained in this work, by Urrutia-Fucugauchi (1981) and the reference direction calculated from the paleopole for stable North America for the period 20 Ma (Besse and Courtillot, 2002). N, number of sites used; Dec, declination; Inc, inclination; α_{95}, 95% confidence limit; k, precision parameter.

Data set	N	Dec °E	Inc (°)	α_{95} (°)	k	Reference
Normal polarity	14	343.8	33.1	11.1	13.02	This work
Reversed polarity	12	173.5	−38.4	10.2	19.00	This work
Combined polarities	26	348.1	35.7	7.4	15.54	This work
Previous data	9	301.2	27.1	10.7		Urrutia-Fucugauchi (1981)
North America Ref. 20 Ma	38	353.0	32.6	2.7		Besse and Courtillot (2002)

significant tectonic movement. We therefore conclude that the paleomagnetic data determined in this work do not suggest any major tectonic movement, when compared with reference data from stable North America.

This result differs from the result obtained previously by Urrutia-Fucugauchi (1981) on the same rock units, who suggested that the study area experienced a large magnitude counterclockwise rotation, ~50°. The corresponding mean direction and paleopole are included in Figs. 8 and 9, for comparison with the reference data and the data obtained in this work. Based on our results, such a rotation seems to be highly unlikely, and we now consider what could have produced the strongly deviating paleomagnetic directions reported by Urrutia-Fucugauchi (1981).

1. The Urrutia-Fucugauchi (1981) study was based on only nine sites, which is not considered to be sufficient for determining a reliable paleodirection or paleopole (e.g., Butler, 1998; Van der Voo, 1993), as it may not have averaged out secular variation. Nevertheless, it would have been extremely accidental to sample rocks either of identical directions during the time variation of the geomagnetic field, or two instances of geomagnetic secular variation with a similar and abnormally extreme field direction during different polarity chrons. The internal coherence of the data and the reasonable dispersion of the data as shown by the confidence circles and angular standard deviation suggest that this is not a feasible explanation.
2. Unreliable determination of ChRM directions. Urrutia-Fucugauchi (1981) did not apply the PCA method as had been recently proposed by Kirschvink (1980) to his demagnetization data. In case of any unresolved multi-component magnetization this could lead to erroneous assignment of ChRM directions. Nevertheless, Urrutia-Fucugauchi (1981) showed that for all but one of his sites demagnetization resulted in stable end point directions. This could mean that very high coercivities and laboratory unblocking temperatures, well beyond those used in experimental procedures, could have produced apparent end point directions, which were not correct. Again, the coherency of the reported site mean directions and their approximate antipodal directions argues against such a possibility.
3. Systematic distortion of site mean directions, in particular the declination, as the inclination values of the nine reported site mean directions compare reasonably well with the reference data, may have been resulted by some systematic error. This incorporates errors during orientation of drill cores in the field (orienting device, magnetic compass, correction of magnetic declination, etc.), measurements in the laboratory, and undetected errors in the programs used for the analysis of the data. These errors are impossible to be analyzed here, so we only may speculate about this potential explanation.

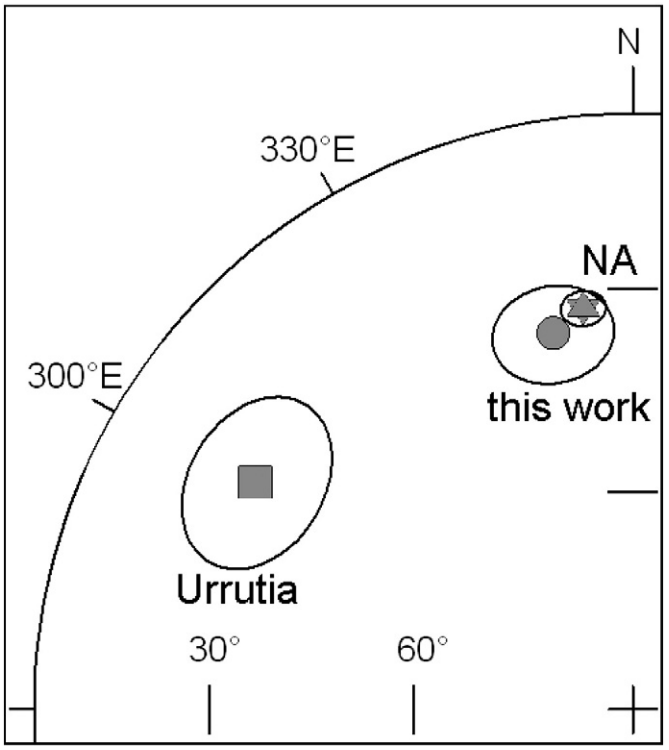


Fig. 8. Overall mean directions with 95% confidence circles compared with the North America reference direction for 20 Ma (star): square, Urrutia-Fucugauchi (1981); dot, this work. Equal area projection, generated with PMag Tools, Version 4.2b.

6. Conclusions

Miocene rocks about 90 km SE of Mexico City have been sampled at 31 sites to assess a previous study by Urrutia-Fucugauchi (1981) involving paleomagnetic data from nine sites from the same study area. The previously obtained mean direction and paleopole are very discordant when compared to the corresponding reference data for stable North America, and Urrutia-Fucugauchi (1981) suggested that the study area had experienced a counter clockwise rotation around a nearby vertical axis, by about 50°. New data obtained in our study are in general from the same types of rocks, the Chalcatzingo trondjhemite domes and Tepexco volcanic rocks, with three additional sites from a nearby intrusion of the same approximate age. These rocks are characterized by the presence of magnetite and often an additional lower Curie temperature phase, identified as high-titanium titanomagnetite. The magnetic granulometry corresponds to pseudo single domain particles. A total of 25 sites provides mean directions of reasonable quality, with most

Table 3
Paleopoles obtained from this work and by Urrutia-Fucugauchi (1981) for the study area, and by Besse and Courtillot (2002) for stable North America, period 20 Ma. N, number of sites used; Long, paleopole longitude; Lat, paleopole latitude; A_{95}, 95% confidence limit. Angular standard deviation s with upper and lower 95% confidence limits s_u and s_l.

Data set	N	Long °E	Lat (°)	A_{95} (°)	k	s (°)	s_u (°)	s_l (°)	Reference
Normal polarities	14	175.5	75.1	10.4		19.8	26.5	15.8	This work
Reversed polarities	12	21.8	−81.5	9.8		17.7	24.2	14.0	This work
Combined polarities	26	184.2	78.3	7.0	17.2	19.0	23.4	16.0	This work
Previous data	9	176.3	33.7	9.5	184.2				Urrutia-Fucugauchi (1981)
North America 20 Ma	38	164.2	83.3	2.7	75.6				Besse and Courtillot (2002)

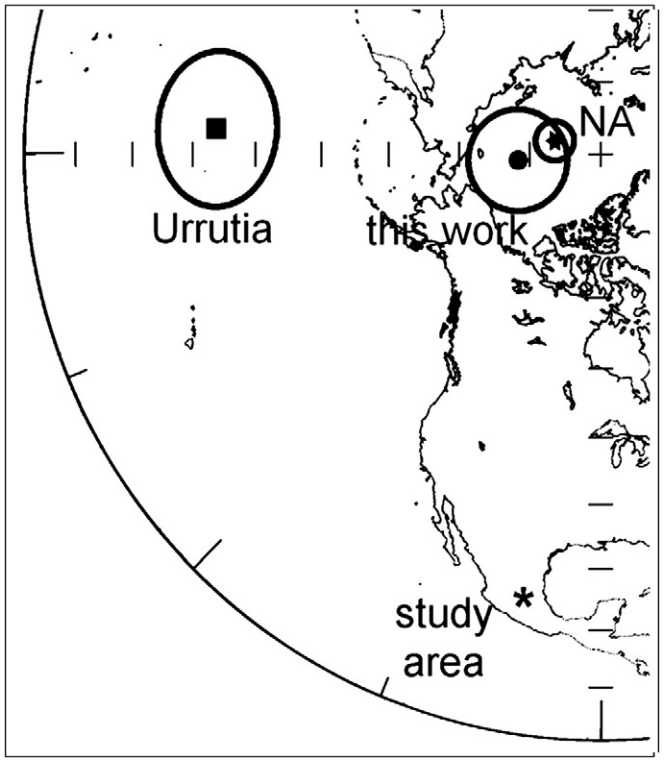


Fig. 9. Paleopoles with 95% confidence circles, compared to the stable North America reference pole NA for 20 Ma (star symbol): square, Urrutia-Fucugauchi (1981); dot, this work; triangle, sampling locality. Equal area projection, generated with PMag Tools, Version 4.2b.

(19) sites having confidence limits $\alpha_{95} < 10°$ and 6 sites with $10.4° < \alpha_{95} < 15.5°$. Both polarities are observed, with 14 (12) sites corresponding to normal (reversed) polarity, which indicates that these rocks cooled over a time period involving at least two polarity chrons. Mean directions for normal and reverse polarities are antipodal and pass a reversal test, with an observed Gamma angle of 9.5° smaller than the 95% critical Gamma = 14.7°. The angular standard deviation of VGPs of s = 19.0° is slightly larger than the value predicted by PSV model G and suggests together with the reversal test that secular variation was sufficiently averaged..An overall mean direction of D = 348.1°E, I = 35.7° with a 95% uncertainty of $\alpha_{95} = 7.4°$ was obtained, which compares well with the reference direction for stable North America for 20 Ma, D = 353.0°, I = 32.6°. Apparent rotation and flattening values are $R = -4.9° \pm 10.4°$ and $F = -2.9° \pm 7.9°$ and thus do not suggest any significant tectonic movement of the study area. Similarly, the paleopole defined by these data (Lat = 78.3°N, Long = 184.2°E, $A_{95} = 7.0°$) compares well with the 20 Ma North America reference pole at Lat = 83.3°N, Long = 164.2°E, $A_{95} = 2.7°$. Apparent tectonic movements calculated from the observed poles define an apparent rotation of $R = -5.3° \pm 6.4°$ and a poleward displacement of $-3.2° \pm 6.0°$, both insignificant within the error limits. Our new data suggest that the study area has not undergone any tectonic movement as suggested in a much earlier study. Because of the much larger number of sites used in our work, their spatial distribution, the magnetic characteristics of the studied rocks, their demagnetization behavior, and a positive reversal test, we therefore propose this result to supersede that of Urrutia-Fucugauchi (1981). Although the previous result was interesting because of the controversial interpretation of a young, large magnitude crustal block rotation without clear accompanying geological evidence, our re-evaluation results in a concordant paleomagnetic result and refutes this interesting, although uncomfortable, inconsistency.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.tecto.2015.07.014.

References

Aguirre-Díaz, G.J., Aranda-Gómez, J.J., Carrasco-Núñez, G., Ferrari, L., 1997. Magmatism and tectonics of Central and Northern Mexico: a selection of the 1997 IAVCEI General Assembly Excursions. Excursion 1. Universidad Nacional Autónoma de México, Instituto de Geofísica, Mexico, D.F., pp. 1–39.

Aranda-Gómez, J.J., Mcdowell, F.W., 1998. Paleogene extension in the southern Basin and Range province of Mexico: syndepositional tilting of Eocene red beds and Oligocene volcanic rocks in the Guanajuato Mining District. Int. Geol. Rev. 40 (2), 116–134.

Ballard, M.M., Van der Voo, R., Urrutia-Fucugauchi, J., 1989. Paleomagnetic results from Grenvillian-aged rocks from Oaxaca, Mexico: evidence for a displaced terrane. Precambrian Res. 42 (3), 343–352.

Beck Jr., M.E., 1980. Paleomagnetic record of plate-margin tectonic processes along the western edge of North America. J. Geophys. Res. 85, 7115–7131.

Besse, J., Courtillot, V., 2002. Apparent and true polar wander and the geometry of the magnetic field in the last 200 years. J. Geophys. Res. 107. http://dx.doi.org/10.1029/2000JB000050.

Böhnel, H., Alva-Valdivia, L., Gonzalez-Huesca, S., Urrutia-Fucugauchi, J., Moran-Zenteno, D.J., Schaaf, P., 1989. Paleomagnetic data and the accretion of the Guerrero terrane, southern México continental margin. In: Hillhouse, J.W. (Ed.), Deep structure and past kinematics of accreted terranes. A.G.U. Geophysical Monograph/I.U.G.G. 5, pp. 73–92.

Böhnel, H., Delgado-Argote, L.A., Kimbrough, D.L., 2002. Discordant paleomagnetic data for Middle-Cretaceous intrusive rocks from northern Baja California: latitude displacement, tilt, or vertical axis rotation? Tectonics 21 (5), 13-1.

Butler, R., 1998. Paleomagnetism: Magnetic Domains to Geologic Terranes. Electronic edition (237 pp.).

Cerca, M., Ferrari, L., López-Martínez, M., Martiny, B., Iriondo, A., 2007. Late Cretaceous shortening and Early Tertiary shearing in the central Sierra Madre del Sur, southern Mexico: insights into the evolution of the Caribbean–North American plate interaction. Tectonics 26 (3), TC3007. http://dx.doi.org/10.1029/2006TC001981.

Collinson, D.W., Runcorn, S.K., 1960. Polar wandering and continental drift: evidence from paleomagnetic observations in the United States. Geol. Soc. Am. Bull. 71, 915–958.

Creer, K.M., Irving, E., Nairn, A.E.M., 1959. Palaeomagnetism of the Great Whin Sill. Geophys. J. Int. 2 (4), 306–323.

Demarest, H.H., 1983. Error analysis of the determination of tectonic rotation from paleomagnetic data. J. Geophys. Res. 88, 4321–4328.

Engebretson, D.C., Cox, A., Gordon, R.G., 1985. Relative motions between oceanic and continental plates in the Pacific basin. Geol. Soc. Am. Spec. Pap. 206, 1–60.

Ferrari, L., López-Martínez, M., Aguirre-Díaz, G., Carrasco-Núñez, G., 1999. Space-time patterns of Cenozoic arc volcanism in central Mexico: from the Sierra Madre Occidental to the Mexican Volcanic Belt. Geology 27, 303–306.

Fisher, R., 1953. Dispersion on a sphere. Proc. R. Soc. Lond. A 217, 295–305.

Fries, C., 1966. Resumen de la Geología de la Hoja Cuernavaca, Estado de Morelos, Guerrero y Puebla: México, D.F., UNAM, Instituto de Gelogía, Carta Geológica de México, Serie de 1:100,000, resumen en el anteverso del mapa.

Godínez-Urban, A., Garza, R.S.M., Geissman, J.W., Wawrzyniec, T., 2011. Paleomagnetism of the Todos Santos and La Silla formations, Chiapas: implications for the opening of the Gulf of Mexico. Geosphere 7 (1), 145–158.

Gómez-Tuena, A., Mori, L., Rincón-Herrera, N., Ortega-Gutiérrez, F., Solé, J., 2008. The origin of a primitive Trondhjemite from the Transmexican Volcanic Belt and its implications for the construction of a modern continental arc. Geology 36, 471–474.

Guzmán-Speziale, M., 2001. Active seismic deformation in the grabens of northern Central America and its relationship to the relative motion of the North America–Caribbean plate boundary. Tectonophysics 337 (1), 39–51.

Henry, C.D., Aranda-Gomez, J.J., 1992. The real southern Basin and Range: mid-to late Cenozoic extension in Mexico. Geology 20 (8), 701–704.

Irving, E., Green, R., 1958. Polar movement relative to Australia. Geophys. J. Int. 1 (1), 64–72.

Irving, E., Wynne, P.J., Thorkelson, D.J., Schiarizza, P., 1996. Large (1000–4000 km) northward movements of tectonic domains in the northern Cordillera, 83 to 45 Ma. J. Geophys. Res. 101, 17901–17916.

Johnson, C.A., Harrison, C.G.A., 1990. Neotectonics in central Mexico. Phys. Earth Planet. Inter. 64 (2), 187–210.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophys. J. Int. 62, 699–718.

Kissel, C., Laj, C., 1988. The Tertiary geodynamical evolution of the Aegean arc: a paleomagnetic reconstruction. Tectonophysics 146 (1), 183–201.

Lenhardt, N., Böhnel, H., Wemmer, K., Torres-Alvarado, I.S., Hornung, J., Hinderer, M., 2010. Petrology, magnetostratigraphy and geochronology of the Miocene volcaniclastic Tepoztlán Formation: implications for the initiation of the Transmexican Volcanic Belt (central Mexico). Bull. Volcanol. 72, 817–832.
Luyendyk, B.P., Kamerling, M.J., Terres, R., 1980. Geometric model for Neogene crustal rotations in southern California. Geol. Soc. Am. Bull. 91, 211–217.
McFadden, P.L., Merrill, R.T., McElhinny, M.W., 1988. Dipole/quadrupole family modeling of paleosecular variation. J. Geophys. Res. 93, 11583–11588.
Mejia, V., Böhnel, H., Opdyke, N.D., Ortega-Rivera, M.A., Lee, J.K.W., Aranda-Gomez, J.J., 2005. Paleosecular variation and time-averaged field recorded in late Pliocene–Holocene lava flows from Mexico. Geochem. Geophys. Geosyst. 6 (7). http://dx.doi.org/10.1029/2004GC000871.
Molina-Garza, R.S., Delgado-Argote, L.A., Böhnel, H., Ramírez, E., Ortega, A., Flores, R.C., 2014. A paleomagnetic transect of the mid-Cretaceous Peninsular Ranges batholith, Baja California, Mexico. Mem. Geol. Soc. Am. 211, 713–737.
Molnar, P., Sykes, L.R., 1969. Tectonics of the Caribbean and Middle America regions from focal mechanisms and seismicity. Geol. Soc. Am. Bull. 80, 1639–1684.
Mooser, F., 1972. The Mexican volcanic belt: structure and tectonics. Geofis. Int. 12 (2), 55–70.
Mooser, F., Nairn, A.E.M., Negendank, J.F.W., 1975. Palaeomagnetic investigations of the Tertiary and Quaternary igneous rocks: VIII. A palaeomagnetic and petrologic study of the Valley of Mexico. Geol. Rundsch. 63, 451–483.
Morán-Zenteno, D.J., Alba-Aldave, L.A., Solé, J., Iriondo, A., 2004. A major resurgent caldera in southern Mexico: the source of the late Eocene Tilzapotla ignimbrite. J. Volcanol. Geotherm. Res. 136 (1), 97–119.
Morán-Zenteno, D.J., Cerca, M., Keppie, J.D., 2007. The Cenozoic tectonic and magmatic evolution of southwestern Mexico: advances and problems of interpretation. Geol. Soc. Am. Spec. Pap. 422, 71–91.
Ortega-Gutiérrez, F., Elías-Herrera, M., Gómez-Tuena, A., Mori, L., Reyes-Salas, M., Macías-Romo, C., Solari, L.A., 2012. Petrology of high-grade crustal xenoliths in the Chalcatzingo Miocene subvolcanic field, southern Mexico: buried basement of the Guerrero–Morelos platform and tectonostratigraphic implications. Int. Geol. Rev. 54 (14), 1597–1634.
Pardo, M., Suarez, G., 1995. Shape of the subducted Rivera and Cocos plates in southern Mexico: seismic and tectonic implications. J. Geophys. Res. 100, 12357–12373.
Rivera, E., De la Teja Segura, M.A., Miranda, A.H., Lemus, O.B., Motolinea, O.G., Leon, V.A., Moctezuma, M.D.S., 1998. Carta geológico-minera Cuernavaca E14–5: Pachuca, Hidalgo, Consejo de Recursos Minerales, scale 1:250,000, 1 sheet.
Ruiz-Martínez, V.C., Urrutia-Fucugauchi, J., Osete, M.L., 2010. Palaeomagnetism of the Western and Central sectors of the Trans-Mexican volcanic belt—implications for tectonic rotations and palaeosecular variation in the past 11 Ma. Geophys. J. Int. 180 (2), 577–595.
Runcorn, S.K., 1956. Paleomagnetic survey in Arizona and Utah: preliminary results. Geol. Soc. Am. Bull. 67, 301–316.
Sedlock, R.L., Ortega-Gutiérrez, F., Speed, R.C., 1993. Tectonostratigraphic terranes and tectonic evolution of Mexico. Geol. Soc. Am. Spec. Pap. 278, 1–153.
Shurbet, D.H., Cebull, S.E., 1984. Tectonic interpretation of the trans-Mexico volcanic belt. Tectonophysics 101, 159–165.
Suter, M., Quintero-Legorreta, O., López-Martinez, M., Aguirre-Díaz, G., Farrar, E., 1995. The Acambay graben: active intraarc extension in the Trans-Mexican volcanic belt, Mexico. Tectonics 14 (6), 1245–1262.
Suter, M., Martínez, M.L., Legorreta, O.Q., Martínez, M.C., 2001. Quaternary intra-arc extension in the central Trans-Mexican volcanic belt. Geol. Soc. Am. Bull. 113 (6), 693–703.
Urrutia-Fucugauchi, J., 1981. Paleomagnetism of the Miocene Jantetelco Granodiorites and Tepexco Volcanic Group and inferences for crustal block rotations in central México. Tectonophysics 76, 149–168.
Urrutia-Fucugauchi, J., Böhnel, H., 1987. Comments on "Tectonic interpretation of the Trans-Mexico Volcanic Belt" by D.H. Shurbet and S.E. Cebull. Tectonophysics 138, 319–323.
Urrutia-Fucugauchi, J., Böhnel, H., 1988. Tectonics along the Trans-Mexican volcanic belt according to paleomagnetic data. Phys. Earth Planet. Inter. 52, 320–329.
Van der Voo, R., 1993. Paleomagnetism of the Atlantic, Tethys and Iapetus Oceans. University Press, Cambridge (411 pp.).
Vazquez Duarte, A.C., 2010. Paleomagnetismo de rocas miocènicas de Chalcatzingo, Morelos, Mèxico: una revaluaciòn (Thesis (licenciatura)), Facultad de Ciencias de la Tierra, Universidad Autònoma de Nuevo Leòn, Mexico (73 pp.).